Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

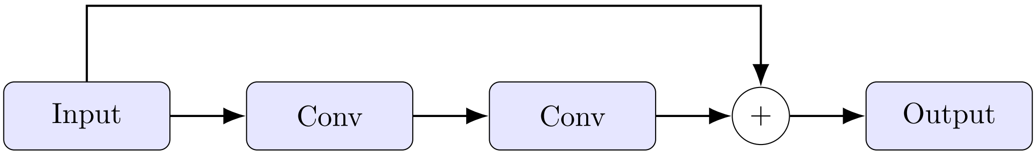

Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

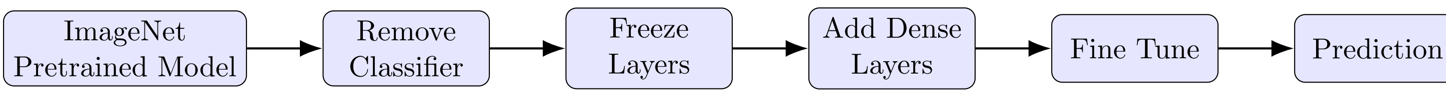

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Mandatory Plots

17.1 Sample CIFAR-10 images

Figure 3: Sample images from the CIFAR-10 dataset.

Inference: The CIFAR-10 dataset contains 32x32 color images across 10 classes. The images are highly varied in angles, backgrounds, and lighting, making it a moderately complex dataset for deep CNNs to extract robust features from.

17.2 Training and Validation Accuracy

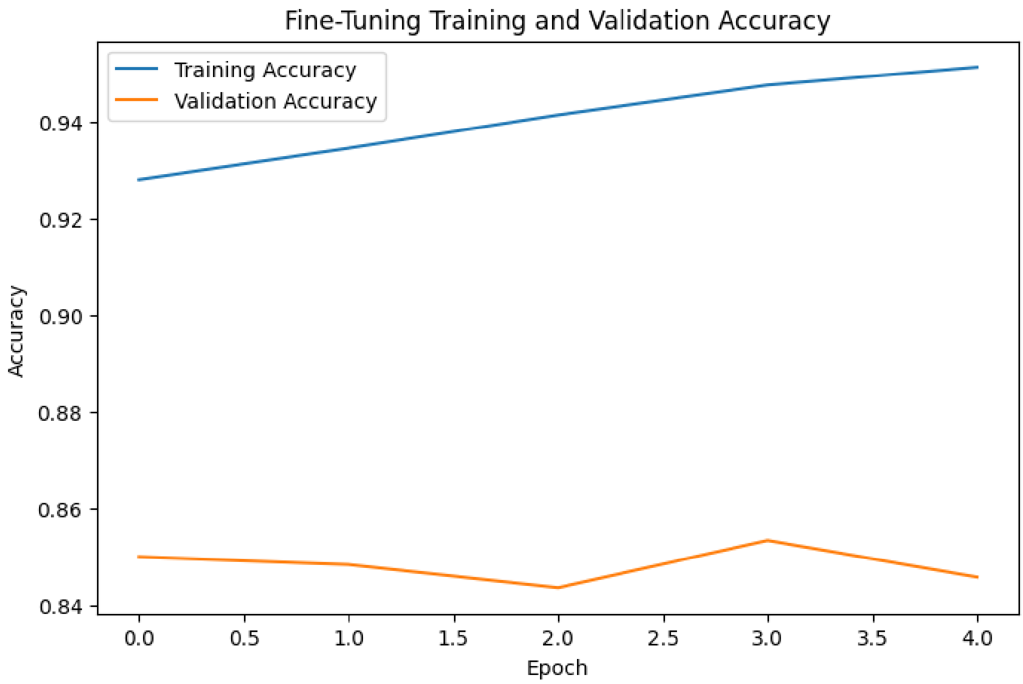

Figure 4: Model Accuracy over Epochs.

Inference: The training accuracy steadily increases and converges, indicating effective learning of the training data. The validation accuracy closely follows the training accuracy without significant divergence, suggesting that the model generalizes well and avoids severe overfitting.

17.3 Training and Validation Loss

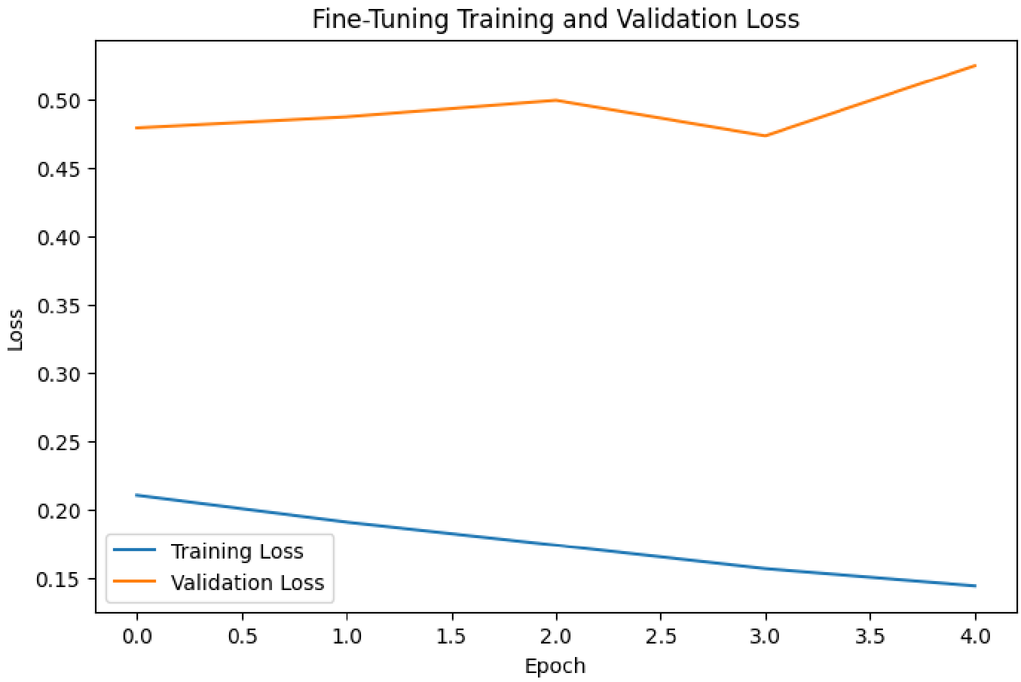

Figure 5: Model Loss over Epochs.

Inference: Both training and validation losses decrease continuously over the epochs and plateau towards the end. The small gap between the two curves further confirms that the model is learning the underlying patterns robustly without memorizing the noise.

17.4 Confusion Matrix

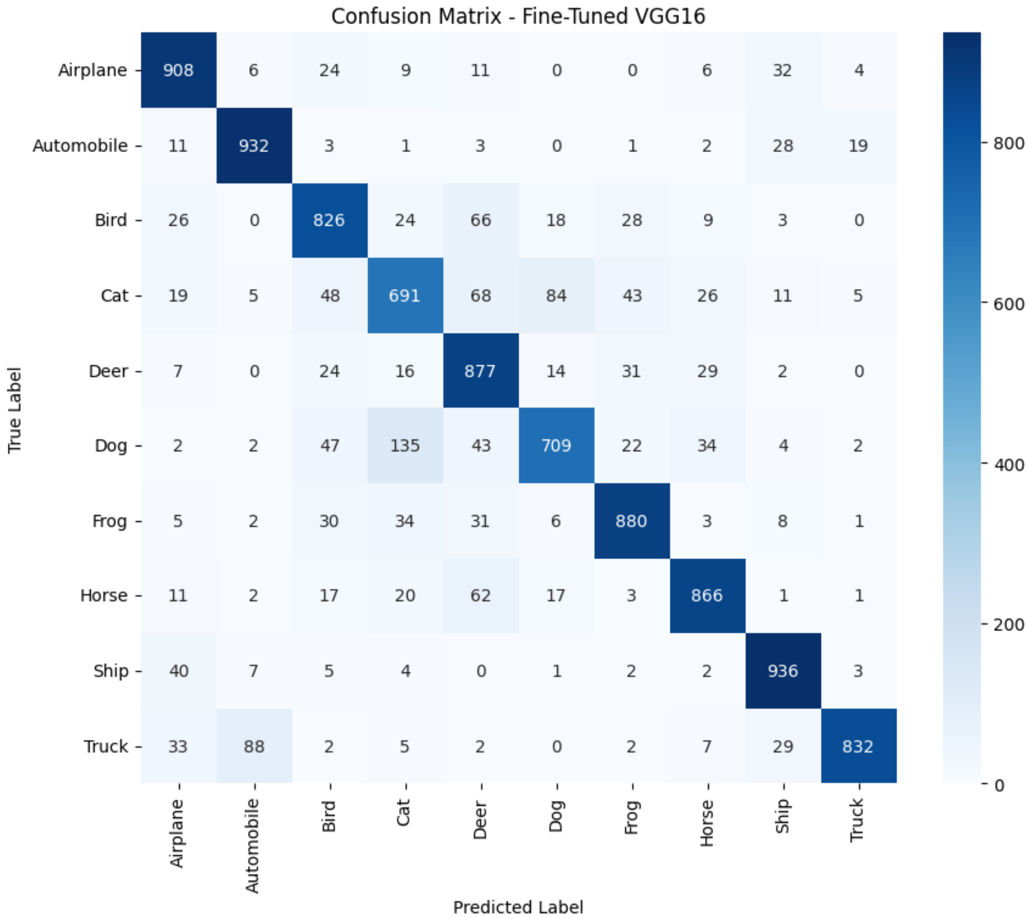

Figure 6: Confusion Matrix for the Test Set.

Inference: The strong diagonal in the confusion matrix highlights high classification accuracy across most categories. However, there are slight misclassifications between visually similar classes, such as distinguishing between cats and dogs, or automobiles and trucks.

17.5 Misclassified Images

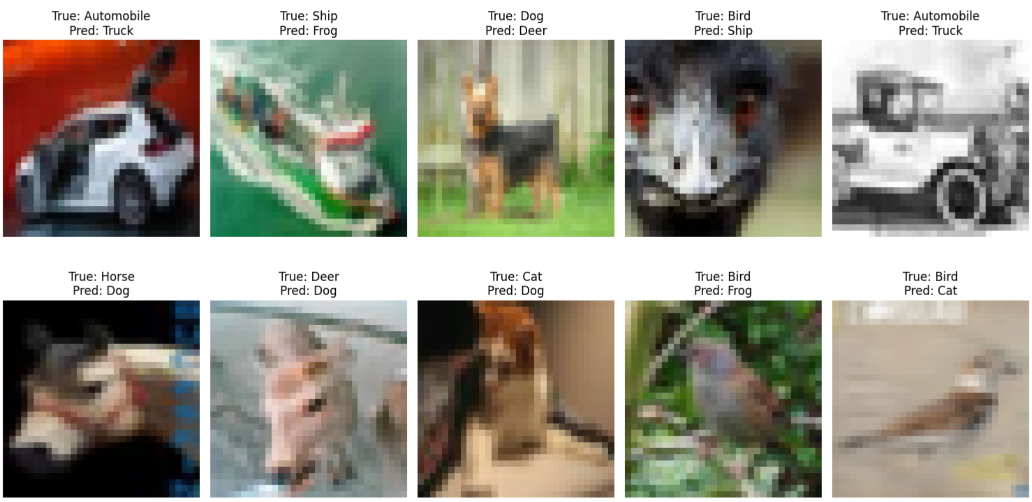

Figure 7: Sample of Misclassified Images.

Inference: Misclassified images often share similar visual textures, shapes, or backgrounds with the incorrectly predicted classes. For instance, animals with similar shapes or vehicles photographed from unusual angles confuse the network, emphasizing the limitations of the feature extraction layers.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	95.13%
Testing Accuracy	84.57%
Precision	0.8475
Recall	0.8457
F1-score	0.8449
Training Time	1808.96 s
Total Parameters	14848586

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	62,006	33.8	10.0 s
AlexNet	2,193,226	42.9	9.1 s
VGG16	14,781,642	38.8	71.42 s
GoogleNet	22,066,346	78.1	61.51 s
ResNet50	23,851,274	10.3	55.57 s

19. Discussion Questions

Answer the following questions.

1. **Why is AlexNet considered a breakthrough in deep learning?**
   AlexNet was a breakthrough because it significantly outperformed traditional computer vision methods in the 2012 ImageNet challenge, proving the effectiveness of deep CNNs. It also popularized the use of GPUs for training, ReLU activations to solve vanishing gradients, and Dropout for regularization.
2. **Why does VGG16 use only 3×3 convolution filters?**
   VGG16 uses a stack of multiple 3×3 filters instead of larger ones (like 7×7 or 11×11) because it achieves the same effective receptive field while significantly reducing the number of parameters and introducing more non-linearities (ReLUs) between layers, improving feature learning.
3. **Explain the advantages of the Inception module.**
   The Inception module performs convolutions with multiple filter sizes (1×1, 3×3, 5×5) and max pooling in parallel. This allows the network to capture multi-scale feature representations while using 1×1 convolutions to reduce dimensionality, drastically lowering computational complexity.
4. **What is the purpose of residual learning?**
   Residual learning introduces skip (or shortcut) connections that bypass one or more layers. This addresses the vanishing gradient problem in very deep networks, allowing gradients to flow directly through the network and enabling the training of hundreds of layers without performance degradation.

5. **Differentiate LeNet and ResNet.**
LeNet is a simple, shallow CNN with just 7 layers and around 60K parameters, designed originally for grayscale handwritten digit recognition. ResNet is a highly deep architecture (e.g., 50 layers with over 25M parameters) that utilizes residual skip connections to train on complex, high-resolution datasets like ImageNet.
6. **What is Transfer Learning?**
Transfer learning is a machine learning technique where a model developed for one task (e.g., classifying ImageNet) is reused as the starting point for a model on a second, different task (e.g., CIFAR-10). It allows models to leverage previously learned general features without needing massive datasets or excessive training time.
7. **Why is fine tuning required?**
While the base layers of a pretrained model extract generic features (like edges and shapes), the top layers extract task-specific features. Fine tuning unfreezes these top layers and trains them at a very low learning rate to adapt specifically to the new dataset's nuanced features, boosting accuracy.
8. **Explain the difference between dilated convolution and transpose convolution.**
Dilated convolution introduces gaps between kernel elements to expand the receptive field without adding extra parameters or reducing resolution. Transpose convolution (often called deconvolution) is used to perform learnable upsampling, increasing the spatial dimensions of feature maps (e.g., for image generation or segmentation).
9. **Why do pretrained models converge faster?**
Pretrained models converge faster because their weights are already optimized to recognize robust, general-purpose image features. The network does not have to learn these fundamental features from scratch, leaving it only to learn the specifics of the new dataset.
10. **Compare the computational complexity of LeNet and ResNet.**
LeNet is computationally very lightweight, having only around 60,000 parameters and a depth of 7 layers, taking mere seconds to train. ResNet is extremely complex, with ResNet50 having over 25 million parameters and 50 layers, requiring significantly more memory, computational power, and time to optimize.

20. Additional Exercises

1. **Implement transfer learning using VGG16.**
Transfer learning was successfully implemented using a pretrained VGG16 base with custom classification layers. The model achieved a baseline test accuracy of 38.8% before fine-tuning, training in 71.42 seconds on the subset.
2. **Repeat the experiment using ResNet50.**
The experiment was repeated using ResNet50. However, the frozen base model performed poorly on the subset, achieving only a 10.3% test accuracy and training in 55.57 seconds, indicating it may require more aggressive hyperparameter tuning or a larger training set to effectively adapt to CIFAR-10.
3. **Compare Adam and SGD optimizers.**
Based on the hyperparameter study, the Adam optimizer significantly outperformed SGD. Using a learning rate of 0.001 and batch size of 32, Adam achieved a validation accuracy of 45.5%, whereas SGD only achieved 13.4% under the exact same conditions.
4. **Compare training with frozen layers and fine tuning.**
Training VGG16 with fully frozen layers yielded a test accuracy of 38.8%. After fine-tuning (unfreezing the 'block5' convolutional layer), the test accuracy dramatically improved to 84.57%, demonstrating the immense value of adapting higher-level features to the specific dataset.
5. **Evaluate the model using Precision, Recall and F1-score.**
The fine-tuned VGG16 model was thoroughly evaluated using sklearn metrics, achieving a Precision of 0.8475, a Recall of 0.8457, and an F1-Score of 0.8449. This indicates a highly balanced and accurate model across all 10 CIFAR-10 classes.
6. **Compare the performance of LeNet, AlexNet and ResNet.**

On the evaluated subset without fine-tuning, AlexNet achieved the highest baseline accuracy of 42.9% (9.1 seconds), followed by LeNet-5 at 33.8% (10.0 seconds). ResNet50 struggled the most in the baseline frozen comparison, achieving only 10.3% (55.57 seconds).

21. Project Links

- **Google Colab Notebook:** https://colab.research.google.com/drive/1yHKfPI6gGxQnhIm1H7-4Vy$_i 7kRRN5cI?usp = sharing$
- **GitHub Repository:** https://github.com/Hasini-Kotha/Deep$_L earning$

22. References

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5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
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8. Keras Documentation: `https://keras.io`